Aleatoric and Epistemic Uncertainty Disentanglement for GAN-based Calorimeter Simulations

Fernando Ugalde Green[1,⋆][0009−0002−6527−2333], Saúl Calderón Ramírez[2,⋆][0000−0001−9993−4388], and Sergio Arguedas Cuendis[3][0000−0003−4234−7005]

[1] Instituto Tecnológico de Costa Rica, Cartago, Costa Rica
fernando.ugalde.green@cern.ch
[2] Instituto Tecnológico de Costa Rica, Cartago, Costa Rica
sacalderon@itcr.ac.cr
[3] CONARE, San José, Costa Rica
sergio.arguedas.cuendis@cern.ch

Abstract. Fast simulation solutions based on generative models have emerged as a promising alternative to computationally expensive Monte Carlo simulation for high-energy physics. However, the lack of reliable Uncertainty Quantification (UQ) remains a significant challenge for a safe usage of neural network based models in this context. In this work, we investigate the use of Monte Carlo Dropout (MCD) for uncertainty disentanglement in a Generative Adversarial Network (GAN) trained to simulate electromagnetic calorimeter showers conditioned on the energy of incident particles. To disentangle predictive uncertainty, MCD is applied at inference time to separate uncertainty estimates into its aleatoric and epistemic components. Experimental results show that the estimated Aleatoric Uncertainty (AU) strongly correlates with output variance, while Epistemic Uncertainty (EU) strongly correlates with distance from the training distribution, yielding near-perfect Pearson correlation coefficients and perfect Spearman rank correlations. These findings show that the proposed method can reliably distinguish between inherent data randomness and uncertainty arising from limited model knowledge, enabling principled detection of unrealistic predictions. The proposed approach requires no significant architectural modifications or additional training objectives, making it easy to integrate into existing neural-network based simulation pipelines while remaining computationally efficient compared to ensemble-based alternatives.

Keywords: Uncertainty Quantification · Uncertainty Disentanglement · Monte Carlo Dropout · Generative Adversarial Network

⋆ Corresponding authors

1 Introduction

1.1 Fast calorimeter simulations

The European Organization for Nuclear Research (CERN) hosts the world's largest and highest-energy particle collider, the Large Hadron Collider (LHC). This machine consists of a 27 km circumference accelerator housed in a massive underground tunnel. At several key points along the ring, massive, multi-layered detectors capture data from particle collisions, measuring energy and trajectories to facilitate rigorous research in high-energy physics. In this work we will focus on the calorimeters, a detector technology used mainly for particle identification. To simulate these high-energy interactions, researchers leverage specialized physics-based simulation packages, such as Geant4 [4]. However, these frameworks are costly to operate and, in the case of the LHCb experiment, approximately 90% of computing resources are allocated to these simulations, leaving 10% available for data management and processing. Recent documents projecting the CPU needs for experiments such as ATLAS [7], CMS [28], and LHCb [12] prove an ever-increasing need for additional computing resources, partially due to the high percentage destined for detailed Monte Carlo simulations. This bottleneck has motivated research on fast calorimeter simulations based on machine learning frameworks, such as deep learning models, to complement (or even replace) expensive Geant4-based simulations.

The Fast Calorimeter Simulation Challenge 2022 [19] (often referred to simply as CaloChallenge) is the world's first open challenge for fast calorimeter simulations. Participants were tasked with training and validating a parametric generative model to approximate the data distribution of detailed calorimeter simulations. To this end, the challenge featured four datasets of Geant4 simulations for generic calorimeters of increasing voxel (readout cell) count. The challenge received a total of 59 submissions, featuring Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Normalizing Flows (NFs), among other generative architectures. GAN and VAE-based models achieved the fastest generation times, while diffusion models achieved the best output quality. The results of this challenge have since been used as a testbench for fast calorimeter simulation developments.

The LHCb collaboration is exploring generative models for fast simulation of its most computationally expensive to simulate components, namely the Ring-Imaging Cherenkov (RICH) detector and the calorimeter system [23]. For simulations of the RICH detector, the Lamarr [8] framework has been implemented to reduce CPU usage by two orders of magnitude in comparison to Geant4-based simulations, using GANs conditioned on the kinematic characteristics of high-energy particles. For simulations of the calorimeter system, the CaloML [22] initiative is being developed based on the results of CaloChallenge to reduce CPU usage of electromagnetic shower simulations by two orders of magnitude as well, using a conditional VAE that directly predicts spatial and energy profiles rather than generating raw energy deposits.

1.2 Uncertainty Quantification

Despite achieving promising results, deep generative models remain black boxes when it comes to reliability assessment and prediction explicability. In other words, it is not trivial to assess whether an output is trustworthy or if it must be considered a hallucination due to being too dissimilar from a particular target distribution. Uncertainty Quantification (UQ) [1] methods allow researchers to predict this phenomena, becoming an important tool for researchers working with deep learning models in safety-critical domains such as healthcare [9,3], chemical processing [13], biodiversity conservation [24], and high-energy physics research [5].

Several UQ methods have been proposed throughout the years, each with their own advantages and use cases, some even tailored for specific architectures in order to minimize computational footprint. One of the earliest methods to achieve widespread adoption was Bayesian Neural Network (BNN) UQ [17], where a neural network learns a distribution over its parameters, allowing Bayesian approximations to estimate predictive uncertainty. However, BNN UQ is computationally expensive and requires significant modifications to the training pipeline that render it unattractive for already trained models. An alternative to BNN UQ is Monte Carlo Dropout (MCD) [16], where dropout is applied at inference time to simulate the behavior of ensemble modeling, such that the variance of multiple forward passes for the same input yields an approximation for predictive uncertainty. This method has also achieved widespread adoption due to its simplicity to implement. Another popular method, building on the theoretical foundations of MCD, is Deep Ensembles [20], which consists of training an ensemble of models for the same task. Performing inference on these models and computing the variance of their predictions for the same input, yields an equivalent uncertainty estimation to MCD. Although previous research has shown Deep Ensembles to be more reliable than MCD in certain domains [17], the computational overhead of training multiple neural networks for the same task may be too significant to be deemed practical.

UQ has been applied to generative models in the context of high-energy physics. Anderlini et al. [5] implemented three UQ methods to quantify the predictive uncertainty of a Cramér GAN [11] trained to generate the delta log-likelihoods of five particle species conditioned on the kinematic characteristics of particles entering the RICH detector of the LHCb experiment. The UQ methods in question were: MCD, adversarial deep ensembles, and distillation. The authors achieved promising results with all three methods, arriving at a principled framework to quantify the sparsity and reliability of several regions of the phase space using UQ. However, these methods compute predictive uncertainty as an overall score sensitive to several sources of uncertainty.

1.3 Uncertainty Disentanglement

An important task in the context of UQ is uncertainty disentanglement [29,26], which deals with the quantification of individual sources of uncertainty, rather

than computing a single score that may aggregate uncertainty from multiple error modes. Uncertainty disentanglement usually deals with separating (and individually quantifying) Aleatoric Uncertainty (AU) and Epistemic Uncertainty (EU). AU refers to variabilities inherent to the training data and, thus, cannot be reduced. EU, on the other hand, refers to lack of knowledge due to architectural limitations or lack of training data and, thus, can be reduced with an improved architecture, with additional training data, or simply through additional training iterations. Uncertainty disentanglement is a valuable tool to identify the source of particular failures in order to treat them in a principled manner. For example, an output that yields high AU, but low EU, is still trustworthy, whereas an output that yields low AU, but high EU, is not trustworthy, despite being an unambiguous event for the model [29].

Previous studies have applied uncertainty disentanglement for real-world applications. Baur et al. [10] developed a benchmark for evaluating uncertainty disentanglement for multi-label chest X-ray classification, applied on convolutional and transformed-based architectures. The authors evaluated 13 UQ methods, including Evidential Deep Learning, HetClass neural networks, and Deep Deterministic Uncertainty, which they extended to the multi-label setting for their chest X-ray task. Luo et al. [21] introduced UDF-GMA as an UQ framework capable of disentangling predictive uncertainty through bayesian approximations. Their method was evaluated on the task of general movement assessment for early detection of brain dysfunction, achieving positive results with the real-world Pmi-GMA dataset.

In the context of high-energy physics, Villalobos-Gómez et al. [30] implemented an efficient UQ method called Feature Densities (FD) [15,2,25], which approximates the EU of a sample by computing the likelihood of its latent features using pre-computed histograms of a hidden layer, using the Cramér GAN implemented by Anderlini et al. in [5]. The inverse of this likelihood corresponds to the final EU score. The authors achieved a performance comparable to MCD while significantly minimizing execution times. Although highly efficient, FD only captures EU. Additionally, the MCD baseline implemented in the study computes EU by subtracting AU from the total (predictive) uncertainty, requiring both uncertainties to be computed to generate EU.

In this work, we explore uncertainty disentanglement in the particular context of generative models, more specifically, a GAN trained for fast calorimeter simulations. Section 2 dives into the specifics of the proposed method, including dataset description, model architecture, and experimental setup. Section 3 includes the results of the proposed uncertainty disentanglement validation experiments. Finally, section 4 features some closing remarks on the investigation.

2 Methodology

2.1 Dataset

To train the generative model of this work, we used Dataset 2 [14] of CaloChallenge, which consists of 200,000 electron shower simulations generated with

Geant4. Of these 200,000 events, 100,000 were used for training, while the remaining 100,000 were used for model evaluation and uncertainty disentanglement. The input feature $x \in \mathbb{R}$ consists of a single scalar per event, representing the incident particle energy, in MeV. These energies were sampled from a log-uniform distribution ranging from 1 GeV to 1 TeV. The output $y \in \mathbb{R}^{6480}$ consists of the energy deposited in a generic cylindrical calorimeter with a uniform geometry of 45 layers, 16 angular bins (slices), and 9 radial bins (rings), yielding a total of $45 \times 16 \times 9 = 6480$ readout cells. These deposited energies are also stored in MeV.

Due to the high dynamic range of these features, we normalized the input feature (incident energy) using log scaling, followed by min-max scaling, as detailed in the following equations:

$$x' = \ln(x + 1), \tag{1}$$

$$x'' = \frac{2(x' - \min(x'))}{\max(x') - \min(x')} - 1, \tag{2}$$

to bring the energies into the uniform and continuous $[-1, 1]$ range. This preprocessing scheme was also applied to the output features (showers).

2.2 Architecture

A Deep Convolutional GAN (DCGAN) [27] was implemented for the task of fast particle transport. Despite recent advances in deep learning showing that architectures such as diffusion models outperform GANs in terms of output quality, we still decided to implement a DCGAN since fast inference is one of the priorities of this study.

The proposed architecture consists of 2 adversarial models:

- A **generator**, which takes the energy of an incoming particle, and concatenates it to a random vector of 100 values sampled from the uniform distribution $\mathcal{U}(0, 1)$. This 101D input is reshaped as a $1 \times 1 \times 1$ volume of 101 channels, which is then processed by several 3D transposed convolution blocks, followed by a final 3D transposed convolution and a hard tanh activation to guarantee bounded outputs, which represent the final electron shower. Each 3D transposed convolution block features a batch normalization module for regularization, and a leaky ReLU activation to introduce non-linearities.
- A **critic**, which takes the energy of an incoming particle, expands it as a $45 \times 16 \times 9$ volume, and concatenates it to an electron shower (which may come from Geant4 or the generator) along the channel dimension. This volume is then processed by several 3D convolution blocks, followed by a final 3D convolution to reduce the input to a single scalar value, representing the realism of the input shower. Each 3D convolution block includes a spectral normalization module for regularization, and a leaky ReLU activation to introduce non-linearities.

The specific modules implemented in this study, along with their hyperparameters, are detailed in figure 1.

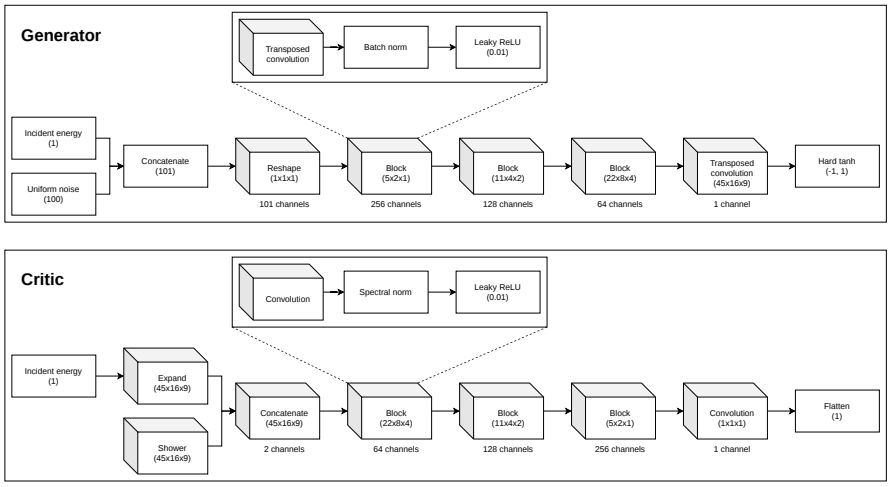


Fig. 1. Architecture of the proposed DCGAN for fast calorimeter simulations.

2.3 Training

Instead of the loss function proposed in the original DCGAN paper, we implemented the loss function of the Wasserstein GAN with Gradient Penalty (WGAN-GP) [18], to avoid the problems associated with the original GAN formulation (vanishing gradients and mode collapse) and the subsequent Wasserstein GAN [6] improvement (exploding gradients). The hyperparameters used in this framework are specified in table 1.

Table 1. Hyperparameters used for model training.

Hyperparameter	Value
Optimizer	Adam
Learning rate	10^{-4}
Betas	(0.5, 0.999)
Epochs	100
Batch size	250
Critic steps	5
Lambda	10

Due to the adversarial nature of GANs, the last epoch will not necesarily yield the best parameters. To circumvent this, we compute the similarity between a sample of generated showers and a sample of truth showers at the end of each epoch. Once training is complete, we save the parameters of the epoch that minimizes this metric, ensuring that we save the best-performing model. The similarity metric in question is the Kolmogorov-Smirnov Distance (KSD), which corresponds to the statistic of the two-sample Kolmogorov-Smirnov test:

$$\mathrm{KSD} = \sup_x |F_1(x) - F_2(x)|, \tag{3}$$

where F_1 and F_2 are the empirical distribution functions of two samples.

2.4 Uncertainty Disentanglement

To disentangle predictive uncertainty into its aleatoric and epistemic components, we propose an extension of MCD that directly outputs uncertainty scores for both uncertainty sources, based upon the work developed in [30]. AU is computed as the variance of n forward passes with dropout disabled:

$$\mathrm{AU}(x) = \mathrm{Var}[G_\theta(x, z)], \tag{4}$$

using generator G with parameters θ. This variance only captures variabilities originating from the random input vector z, thus, limiting the score to data uncertainty. Conversely, EU is computed as the variance of n forward passes with dropout enabled at inference time and with a fixed random input vector z across forward passes:

$$\mathrm{EU}(x) = \mathrm{Var}[G_{\theta'}(x, z_{\mathrm{fixed}})], \tag{5}$$

where θ' is the set of parameters of the model multiplied by a random dropout mask. By fixing z and enabling dropout, we guarantee this variance only captures variabilities in the model parameters, thus, capturing only model uncertainty. To the best of our knowledge, this is the first implementation of a direct EU quantification for GANs through fixed input noise.

2.5 Experimental setup

To assess the performance of the trained model, we evaluate the difference between the true and generated showers over three high-level features: the longitudinal profile (total energy per calorimeter layer), the radial profile (total energy per calorimeter ring), and the total energy per shower. To this end, we generate showers conditioned on the incident energies of the validation dataset, and compute the high-level features for both the generated showers and the true showers in the validation dataset. For each of these features, we compute the normalized difference between the real and generated results. Ideally, this difference should be as close to 0% as possible, signaling that generated showers follow a distribution close to the true showers.

To validate the proposed uncertainty disentanglement method, we compute both AU and EU for generated showers conditioned on the incident energies of the validation dataset. We split the showers in equally-sized sets of increasing uncertainty, such that the showers of the first set have the lowest uncertainty scores, while the showers of the last set carry the highest uncertainty scores. We correlate the mean uncertainty of each of these disjoint sets with an appropriate metric that reflects the expected behavior under the computed uncertainty scores, which we detail later. The correlation metrics used are the Pearson Correlation Coefficient (PCC) and Spearman's rank correlation coefficient.

PCC measures the linear and monotonic relationship between two random variables X and Y as follows:

$$\rho_{X,Y} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}, \tag{6}$$

while Spearman's rank correlation coefficient (Spearman's ρ), solely quantifies monotonic relationships between those variables, regardless of linearity. Spearman's ρ reduces to computing the PCC between the rank values of X and Y.

To validate AU, we compute the variance of the true showers in each set, and correlate these variances to the mean AU of each set, expecting a strong positive correlation (the higher the AU, the higher the variance of the output features should be). To validate EU, we compute the KSD between the real and generated showers in each set, and correlate these distances to the mean EU of each set, expecting a strong positive correlation (the higher the EU, the more dissimilar the outputs should be with respect to the true data distribution). These experiments are repeated 10 times with different batches of true data, and the final mean and standard deviation are reported.

The hyperparameters used for both experiments are specified in table 2.

Table 2. Hyperparameters used for uncertainty validation.

Hyperparameter	Value
Number of forward passes (n)	100
Dropout probability (p)	0.1
Number of disjoint sets	10
Experiment repetitions	10

As it is important in real-world UQ to use a reasonable amount of computing resources, we measure the computing time of both model inference and UQ. To this end, we conduct a final test which involves measuring the time it takes the model to generate showers and perform UQ. We first generate four batches of input data with increasing sample sizes: 1, 10, 100, and 1000 samples. We perform inference on these batches and record shower generation and UQ timings. Finally, we take the average and standard deviation of 10 sequential runs and report the results.

3 Results

Figure 2 shows the evolution of the KSD between real and generated samples over the 100 training epochs. After training for a total duration of 5.53 hours, this metric was minimized at epoch 95 with a value of 0.0575. Figure 3 allows for a qualitative inspection of a generated shower against a true shower.

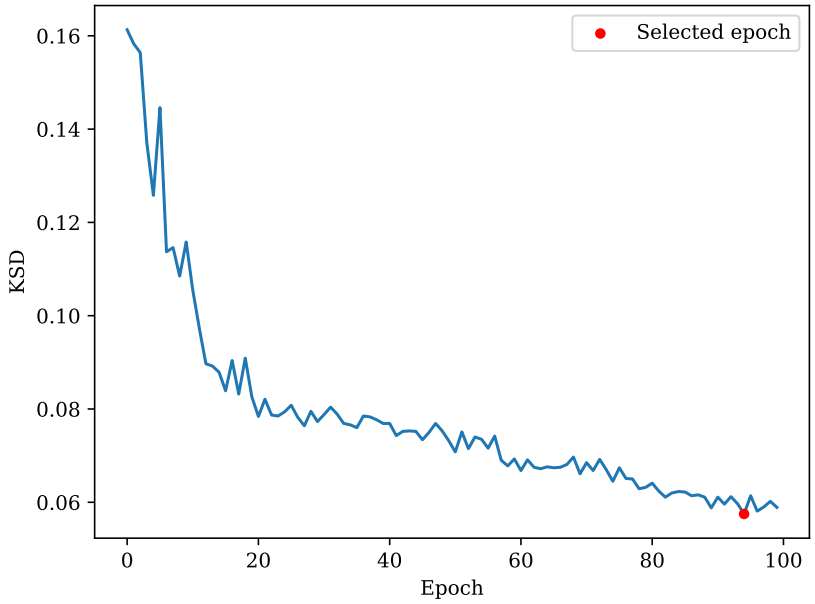


Fig. 2. Evolution of the KSD between real and generated samples over 100 training epochs of the proposed WGAN-GP. The red dot marks the best performing epoch.

Figure 4 shows the resulting high-level features for both real and generated datasets, along with their normalized differences. The longitudinal and radial profiles yielded, respectively, a mean difference of $0.7\% \pm 9.7\%$ and $-4.02\% \pm 3.73\%$ between real and generated results. The total energy per shower, split across 100 bins, resulted in a mean difference of $-2.75\% \pm 7.35\%$ for the first (most significant) 95 bins of each histogram. Although these metrics could be further improved with additional training iterations, we consider the model sufficiently good to proceed with uncertainty disentanglement validation analyses.

Figure 5 shows the results of the uncertainty disentanglement experiments for both AU and EU. This figure was generated using the data points featured in tables 5 and 6. As a final summary, table 3 features the final correlation coefficients achieved over the 10 experimental runs.

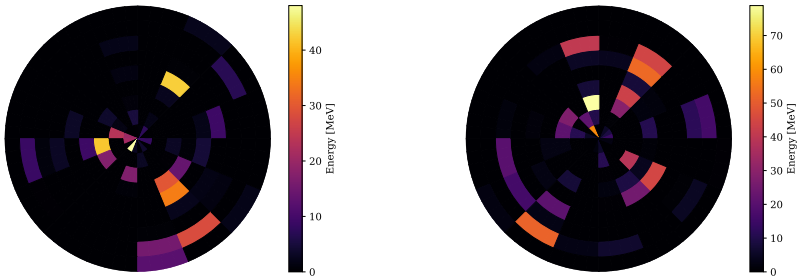


Fig. 3. A true shower from the training dataset (left) and a generated shower (right). The voxels shown in each plot correspond to the 144 readout cells of layer 0.

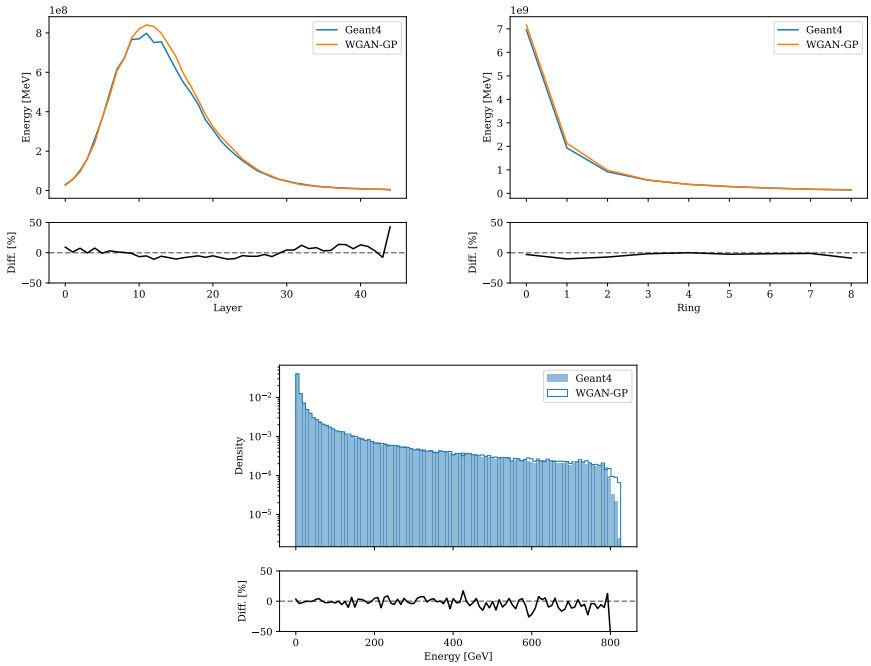


Fig. 4. Longitudinal profile (top left), radial profile (top right) and total energy per shower (bottom), for both real and generated datasets, along with their respective normalized differences.

Table 3. Correlation coefficients for both AU and EU validation experiments.

Experiment	PCC	Spearman's ρ
AU vs. Variance	0.9908 ± 0.0009	1.0000 ± 0.0000
EU vs. KSD	0.9707 ± 0.0025	1.0000 ± 0.0000

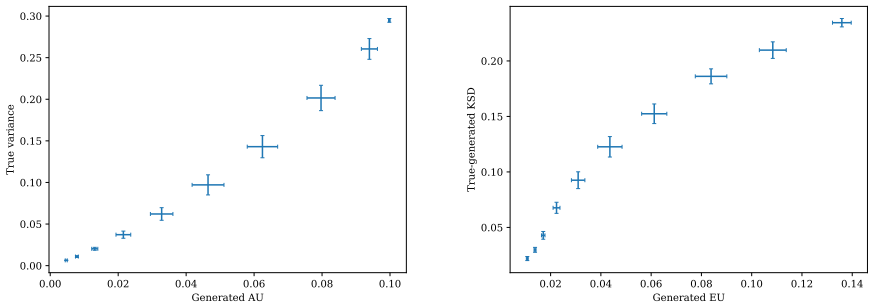


Fig. 5. Correlation curves for AU validation (left) and EU validation (right). Each point corresponds to a disjoint set of showers, while the error bars themselves represent one standard deviation in each axis.

Finally, table 4 features execution times for both shower generation and UQ, over batches of increasing sample size. These timings were generated on a 32 GB NVIDIA Tesla V100 GPU.

Table 4. Execution times for shower generation and UQ.

Batch size	Generation time [s]	UQ time [s]
1	$5.16 \times 10^{-4} \pm 1.70 \times 10^{-4}$	0.083 ± 0.002
10	$4.69 \times 10^{-4} \pm 8.39 \times 10^{-5}$	0.087 ± 0.003
100	$4.47 \times 10^{-4} \pm 3.63 \times 10^{-5}$	0.575 ± 0.029
1000	$4.38 \times 10^{-4} \pm 3.21 \times 10^{-5}$	5.695 ± 0.292

4 Discussion

This work shows the performance of an MCD-based uncertainty disentanglement method applied on a GAN trained for calorimeter simulations. The GAN is capable of effectively simulating electromagnetic showers conditioned on the energy of incident particles. The generated showers exhibit no sign of mode collapse, and are capable of reconstructing high-level features, such as the total energy per calorimeter layer. Overall model performance could be further improved with an alternative architecture or simply with additional training steps. However, we did not focus on this task since the objective of this work is uncertainty disentanglement and not pure model reliability.

The proposed uncertainty disentanglement method effectively separates predictive uncertainty into its aleatoric and epistemic components. The proposed experiments yielded near perfect correlations with PCC and perfect correlations

with Spearman's ρ. This means that AU scores effectively correlate with output variances, while EU scores effectively correlate with the distance between the generated and real data distributions, allowing principled detection of unrealistic predictions.

The timing results indicate that the shower generation process is highly efficient, requiring approximately 4.68×10^{-4} seconds per batch, independently of sample size. However, UQ introduces additional computational overhead due to the multiple forward passes required by MCD. Despite this increase, the method remains computationally practical, considering that no additional models must be trained or stored, unlike ensemble-based approaches.

Overall, these results prove that MCD can provide meaningful uncertainty estimates without significant architectural modifications or additional training objectives. The method is straightforward to implement, computationally inexpensive compared to ensemble-based approaches, and can be integrated into existing deep learning simulation pipelines. Additionally, by providing disentangled uncertainty estimates, end-users can make informed decisions on how to deal with uncertain outputs depending on the dominating uncertainty source. Future research could explore the applicability of the proposed method to alternative generative architectures and to larger and more diverse datasets. Such developments could improve the reliability and interpretability of deep learning fast simulation frameworks for high-energy physics and other safety-critical domains.

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References

1. Abdar, M., Pourpanah, F., Hussain, S., Rezazadegan, D., Liu, L., Ghavamzadeh, M., Fieguth, P., Cao, X., Khosravi, A., Acharya, U.R., Makarenkov, V., Nahavandi, S.: A review of uncertainty quantification in deep learning: Techniques, applications and challenges. Information Fusion **76**, 243–297 (2021). https://doi.org/https://doi.org/10.1016/j.inffus.2021.05.008, https://www.sciencedirect.com/science/article/pii/S1566253521001081
2. Abreu-Cardenas, M., Calderón-Ramírez, S.: Feature densities for uncertainty quantification for complex text detection in spanish. In: 2025 IEEE 7th International Conference on BioInspired Processing (BIP). pp. 1–6. IEEE (2025)
3. Abreu Cárdenas, M.G., Calderón Ramírez, S., Rojas Gonzalez, S.: Evolutionary multi-objective calibration of mc-dropout predictive uncertainty quantification for diabetic retinopathy. In: Proceedings of the Genetic and Evolutionary Computation Conference Companion. pp. 253–256 (2026)
4. Agostinelli, S., Allison, J., Amako, K.a., Apostolakis, J., Araujo, H., Arce, P., Asai, M., Axen, D., Banerjee, S., Barrand, G., et al.: Geant4—a simulation toolkit. Nuclear instruments and methods in physics research section A: Accelerators, Spectrometers, Detectors and Associated Equipment **506**(3), 250–303 (2003)

5. Anderlini, L., Chimpoesh, C., Kazeev, N., Shishigina, A., collaboration, L.: Generative models uncertainty estimation. In: Journal of Physics: Conference Series. vol. 2438, p. 012088. IOP Publishing (2023)
6. Arjovsky, M., Chintala, S., Bottou, L.: Wasserstein gan (2017), https://arxiv.org/abs/1701.07875
7. ATLAS Collaboration: ATLAS Software and Computing HL-LHC Roadmap. Tech. rep., CERN, Geneva (2022), https://cds.cern.ch/record/2802918
8. Barbetti, M.: Lamarr: Lhcb ultra-fast simulation based on machine learning models deployed within gauss (2024), https://arxiv.org/abs/2303.11428
9. Barquero, B.C., Calderon-Ramirez, S., Rodriguez-Rivas, D.: Guided data augmentation by transfer function using uncertainty scores for medical image classification. In: 2024 IEEE 42nd Central America and Panama Convention (CONCAPAN XLII). pp. 1–6. IEEE (2024)
10. Baur, S., Samek, W., Ma, J.: Benchmarking uncertainty and its disentanglement in multi-label chest x-ray classification. In: International Workshop on Uncertainty for Safe Utilization of Machine Learning in Medical Imaging. pp. 193–203. Springer (2025)
11. Bellemare, M.G., Danihelka, I., Dabney, W., Mohamed, S., Lakshminarayanan, B., Hoyer, S., Munos, R.: The cramer distance as a solution to biased wasserstein gradients. arXiv preprint arXiv:1705.10743 (2017)
12. Bozzi, C.: LHCb Computing Resources: 2025 requests. Tech. rep., CERN, Geneva (2024), https://cds.cern.ch/record/2888939
13. Dokas, I.M.: From hallucinations to hazards: benchmarking llms for hazard analysis in safety-critical systems. Safety Science **194**, 107056 (2026). https://doi.org/https://doi.org/10.1016/j.ssci.2025.107056, https://www.sciencedirect.com/science/article/pii/S0925753525002814
14. Faucci Giannelli, M., Kasieczka, G., Krause, C., Nachman, B., Salamani, D., Shih, D., Zaborowska, A.: Fast calorimeter simulation challenge 2022 - dataset 2 (Mar 2022). https://doi.org/10.5281/zenodo.6366271, https://doi.org/10.5281/zenodo.6366271
15. Fuentes-Fino, R., Calderón-Ramírez, S., Domínguez, E., López-Rubio, E., Elizondo, D., Molina-Cabello, M.A.: An uncertainty estimator method based on the application of feature density to classify mammograms for breast cancer detection. Neural Computing and Applications **35**(30), 22151–22161 (2023)
16. Gal, Y., Ghahramani, Z.: Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In: international conference on machine learning. pp. 1050–1059. PMLR (2016)
17. Gawlikowski, J., Tassi, C.R.N., Ali, M., Lee, J., Humt, M., Feng, J., Kruspe, A., Triebel, R., Jung, P., Roscher, R., et al.: A survey of uncertainty in deep neural networks. Artificial intelligence review **56**(Suppl 1), 1513–1589 (2023)
18. Gulrajani, I., Ahmed, F., Arjovsky, M., Dumoulin, V., Courville, A.C.: Improved training of wasserstein gans. Advances in neural information processing systems **30** (2017)
19. Krause, C., Faucci Giannelli, M., Kasieczka, G., Nachman, B., Salamani, D., Shih, D., Zaborowska, A., Amram, O., Borras, K., Buckley, M.R., et al.: Calochallenge 2022: a community challenge for fast calorimeter simulation. Reports on Progress in Physics **88**(11), 116201 (2025)
20. Lakshminarayanan, B., Pritzel, A., Blundell, C.: Simple and scalable predictive uncertainty estimation using deep ensembles. Advances in neural information processing systems **30** (2017)

21. Luo, Z., Gooya, A., Ho, E.S.: Udf-gma: Uncertainty disentanglement and fusion for general movement assessment. IEEE Journal of Biomedical and Health Informatics (2025)
22. Mazurek, M.: New simulation software and machine learning technologies in the lhcb experiment to evaluate physics performance of run 3 (May 2024)
23. Mazurek, M.: Machine learning in LHCb Simulation : From fast to flash. PoS **LHCP2025**, 125 (2025). https://doi.org/10.22323/1.499.0125, https://cds.cern.ch/record/2945115, proceedings for The Thirteenth Annual Large Hadron Collider Physics (LHCP2025), 5-9 May 2025, Taipei
24. Mora-Cross, M., Calderon-Ramirez, S.: Uncertainty estimation in large language models to support biodiversity conservation. In: Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 6: Industry Track). pp. 368–378 (2024)
25. Mora-Cross, M., Calderón-Ramírez, S.: Uncertainty quantification in large language models using feature space density and clustering. In: 2025 IEEE 7th International Conference on BioInspired Processing (BIP). pp. 1–7. IEEE (2025)
26. Mucsányi, B., Kirchhof, M., Oh, S.J.: Benchmarking uncertainty disentanglement: Specialized uncertainties for specialized tasks. Advances in neural information processing systems **37**, 50972–51038 (2024)
27. Radford, A., Metz, L., Chintala, S.: Unsupervised representation learning with deep convolutional generative adversarial networks. arXiv preprint arXiv:1511.06434 (2015)
28. Software, C.O., Computing: CMS Phase-2 Computing Model: Update Document. Tech. rep., CERN, Geneva (2022), https://cds.cern.ch/record/2815292
29. Valdenegro-Toro, M., Mori, D.S.: A deeper look into aleatoric and epistemic uncertainty disentanglement. In: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW). pp. 1508–1516. IEEE (2022)
30. Villalobos-Gómez, E., Ugalde Green, F., Arce Morales, J., Pastor Barrientos, J.D., Anderlini, L., Calderon-Ramirez, S., Arguedas Cuendis, S.: Uncertainty quantification for a cramér generative adversarial network-based simulation of the ring-imaging cherenkov particle detector using feature densities. EPJ Research Infrastructures **10**(1), 15 (2026)

A Detailed results

Table 5. Generated AU and true variance for 10 disjoint sets of the validation dataset. The reported means and standard deviations were generated from 10 experiment repetitions.

Set index	Generated AU	True variance
1	0.0047 ± 0.0003	0.0064 ± 0.0003
2	0.0079 ± 0.0004	0.0109 ± 0.0011
3	0.0131 ± 0.0009	0.0202 ± 0.0018
4	0.0215 ± 0.0022	0.0372 ± 0.0043
5	0.0328 ± 0.0033	0.0621 ± 0.0076
6	0.0464 ± 0.0047	0.0971 ± 0.0121
7	0.0625 ± 0.0045	0.1430 ± 0.0133
8	0.0797 ± 0.0041	0.2016 ± 0.0152
9	0.0939 ± 0.0024	0.2605 ± 0.0125
10	0.0998 ± 0.0002	0.2948 ± 0.0023

Table 6. Generated EU and KSD between real and generated data, for 10 disjoint sets of the validation dataset. The reported means and standard deviations were generated from 10 experiment repetitions.

Set index	Generated EU	True-generated KSD
1	0.0107 ± 0.0004	0.0220 ± 0.0018
2	0.0138 ± 0.0004	0.0298 ± 0.0022
3	0.0171 ± 0.0007	0.0429 ± 0.0035
4	0.0224 ± 0.0014	0.0677 ± 0.0050
5	0.0310 ± 0.0027	0.0925 ± 0.0076
6	0.0436 ± 0.0048	0.1226 ± 0.0092
7	0.0612 ± 0.0050	0.1524 ± 0.0088
8	0.0838 ± 0.0063	0.1861 ± 0.0068
9	0.1084 ± 0.0053	0.2097 ± 0.0075
10	0.1359 ± 0.0037	0.2345 ± 0.0038